

Tribhuvan University Faculty of Humanities and Social Science

A PROJECT PROPOSAL ON

PORTFOLIO MANAGEMENT SYSTEM

Submitted to

Department of Computer Application

Nepal Mega College

In partial fulfillment of the requirements for the Bachelors in Computer Application

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June 2026

Table of Contents

1. Introduction 1
2. Problem Statement 1
3. Objective 1
4. Methodology 2
   A) Requirement Identification 2
      i) Study of existing system 2
      ii) Requirement Collection 2
      iii) Functional Requirement 2
      iv) Non-Functional Requirement 2
   B) Feasibility Study 3
      i) Technical Feasibility 3
      ii) Operational Feasibility 3
      iii) Economic Feasibility 3
   C) High Level Design of System 4
      i) System Flowchart 4
      ii) Flowchart for Admin 5
      iii) Use case diagram 6
      iv) Methodology of the proposed system 7
5) GANTT Chart 8
6) Expected Outcome 8
7) References 9

1. Introduction

The Portfolio Management System is a web-based application designed to help users manage and track their stock investments in a structured and organized manner. The system allows users to simulate multiple demat accounts, store stock holdings, and calculate profit or loss based on recorded investment data. It also includes a market information module that provides IPO updates and market-related announcements [1] [2].

This system is designed as an academic project to demonstrate database design, user authentication, and CRUD operations using web technologies without integration with real financial APIs or external trading platforms. [3] [4]

2. Problem Statement

- Investor struggle to track multiple investments in an organized way
- No simple system exists for simulating portfolio management for learning purposes.
- Manual calculation of profit/loss is time-consuming and error-prone.
- Market updates like IPOs news are scattered across different sources.

3. Objective

- To build a user authentication system (login/register)
- To allow users to manage multiple demat accounts
- To store and manage stock holdings
- To demonstrate CRUD operations and relational database design
- To generate and manage investment portfolio

4. Methodology

A) Requirement Identification

i) Study of existing system

Existing systems like spreadsheets and financial apps are used to track investments, but they are either too complex or not suitable for learning purposes. Most systems do not provide a simple simulated environment for beginners. There is no single platform combining portfolio tracking and market information for educational use [5].

ii) Requirement Collection

- The system requires a user registration and login module for secure access.
- Discussion with IPO investors and Demat account holders.
- Analysis of existing portfolio management practices
- Review of Nepalese stock market investment workflows

iii) Functional Requirement

- User registration and login
- Manage demat accounts
- Add/update/delete stock holdings
- Calculate profit and loss
- Show IPO market news

iv) Non- Functional Requirement

- User-friendly interface
- Data Security
- Reliability
- Fast system response
- Basic user data security

B) Feasibility Study

i)Technical Feasibility

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes for the implementing this system. [6]

The project can be developed using:

- Frontend: HTML,CSS, JavaScript
- Backend: PHP
- Database: MySQL
- Server Environment: Apache Web Server (Localhost / XAMPP)

The required technologies are readily available and suitable for academic project development.

ii) Operational Feasibility

The system is designed to be simple, intuitive, and user-friendly so that students and beginner users can easily operate it without prior technical or financial knowledge. It provides smooth navigation between different modules such as user management, portfolio tracking, and market information. The system reduces manual work by automating tasks like profit and loss calculation and data storage. It also minimizes errors that usually occur in manual tracking systems. Overall, it ensures efficient and practical use in an academic environment.

iii) Economic Feasibility

The project is economically feasible as it is developed using free and open-source technologies such as HTML, CSS, JavaScript, PHP, and MySQL. No paid software, licenses, or third-party services are required for development. It can be built using basic hardware resources such as a standard personal computer or laptop. The system also does not require any external infrastructure costs for academic deployment. Hence, it is highly cost-effective and suitable for student-level implementation.

C) High Level Design of System

i) System Flowchart

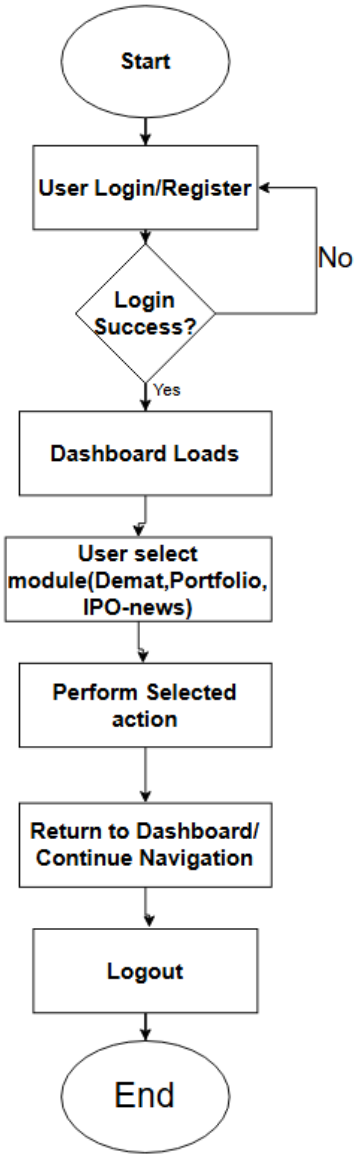


Figure 1: System Flowchart for user

The flowchart shows how a user interacts with the system. The user first opens the website and logs in or registers. The system checks credentials and allows access only if they are valid. After login, the user enters the dashboard with sidebar navigation. From there, the user can access modules like Demat, Portfolio, IPO News, or Logout. After performing actions, the user returns to the dashboard until they choose to logout, which ends the session.

i) Flowchart for Admin

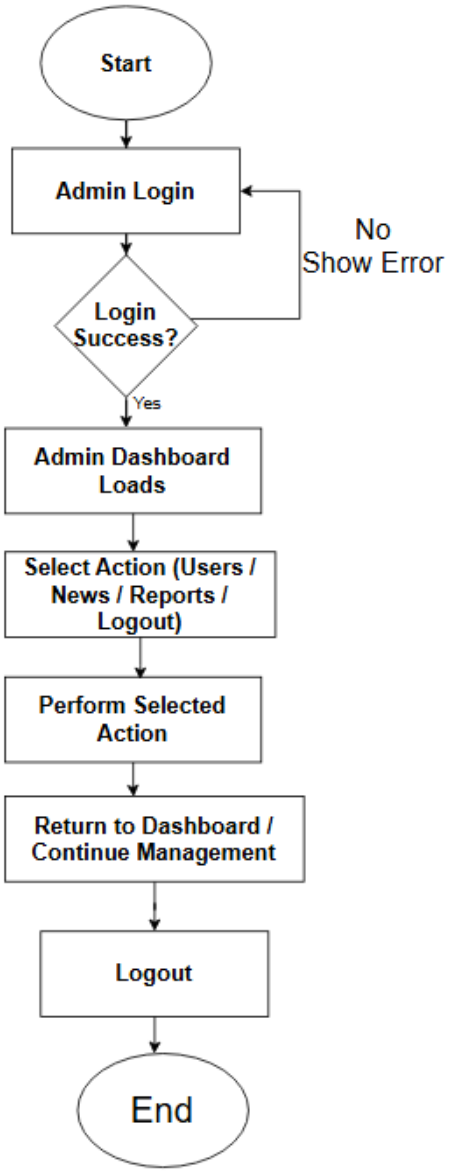


Figure 2: Admin Flowchart

The flowchart shows how the admin accesses and manages the system. The admin logs in and the system validates the credentials. If valid, the admin enters the dashboard. From there, the admin can manage users, update IPO news data, or view reports. After performing actions, the admin returns to the dashboard until they choose to logout, which ends the session.

ii) Use case diagram

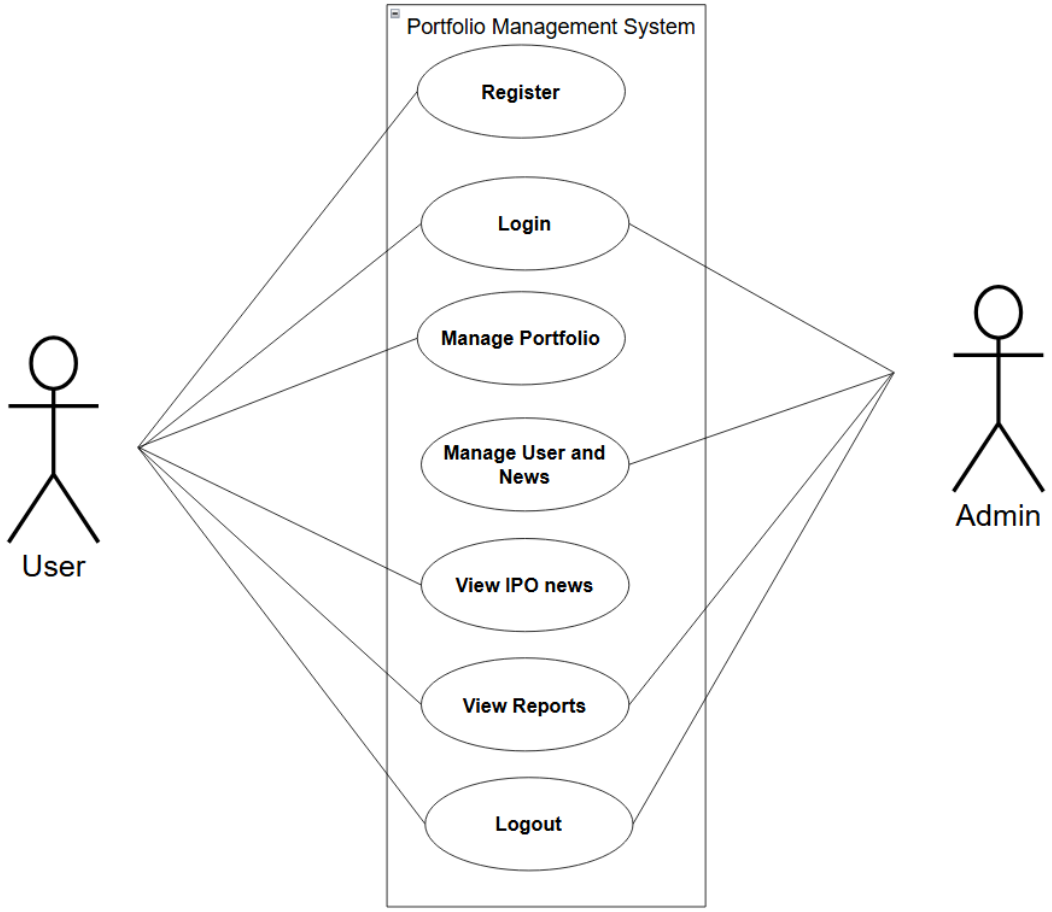


Figure 3: Use Case Diagram for Portfolio Management System

This UML use case diagram maps the core features of a Portfolio Management System for two distinct actors. On the left, a regular User connects to consumer features like account registration, portfolio management, and viewing IPO news. On the right, an Admin connects to high-level privileges including user and news management alongside system report viewing. Both actors are mapped to their respective login use cases to gain secure entry into the application boundary. Finally, the lines from both the User and Admin converge at a single, shared Logout use case at the bottom. This layout cleanly separates standard user capabilities from backend administrative controls. [6]

iii) Methodology of the proposed system

Waterfall Model

The Waterfall Model is a traditional software development methodology that follows a linear, phase-by-phase approach, where each phase must be completed before moving to the next. It does not allow backtracking and permits only minimal changes once a phase is completed. The model is simple, well-structured, and easy to manage, offering high predictability and clearly defined milestones throughout the development process. [6]

Why Waterfall Model?

We use the Waterfall Model for this project because the requirements such as login system, portfolio management, IPO news module, and admin panel are clearly defined from the beginning. It follows a simple and structured step-by-step development approach, which makes it easier to manage and implement each module separately. This model is suitable for database-driven academic web applications like the Portfolio Management System.

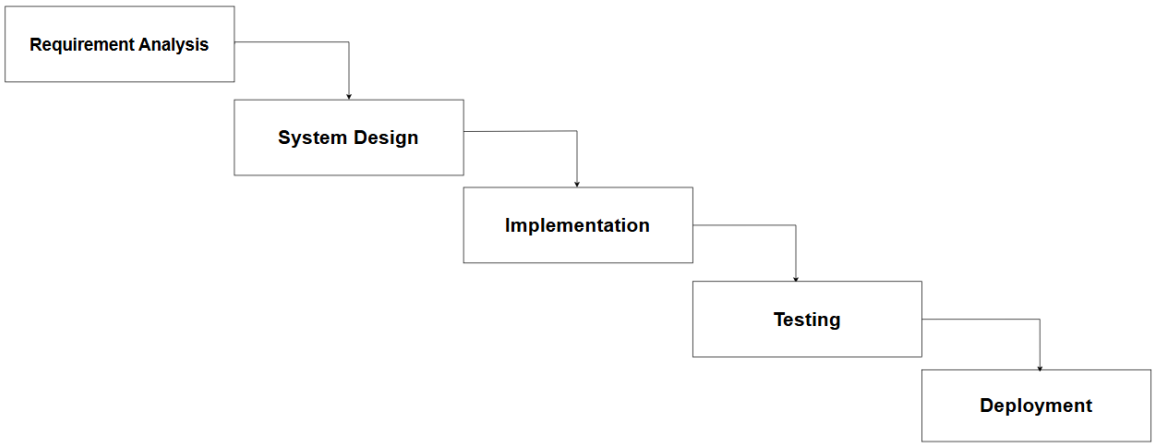


Figure 4: Waterfall Model for Portfolio Management System

This figure represents the Waterfall software development model used in the project. It shows the sequential phases including Requirement Analysis, System Design, Implementation, Testing, and Deployment. Each phase is completed step by step to ensure a structured development process of the Portfolio Management System.

5) GANTT Chart

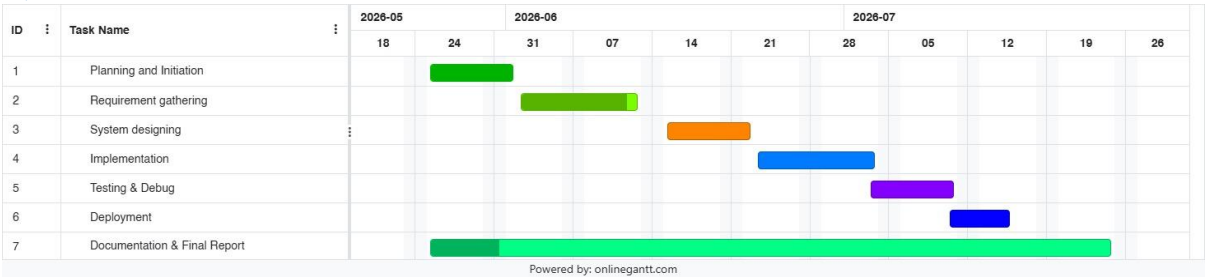


Figure 5: GANTT chart for Portfolio Management System

This Gantt chart illustrates the project schedule and timeline for developing the Portfolio Management System. The project begins with planning and requirement gathering, followed by system design and implementation. After development, testing and debugging are conducted to identify and fix errors before deployment. Documentation and report preparation are carried out throughout the project and completed before the final submission. The chart helps visualize the duration and sequence of activities from project initiation to completion.

6) Expected Outcome

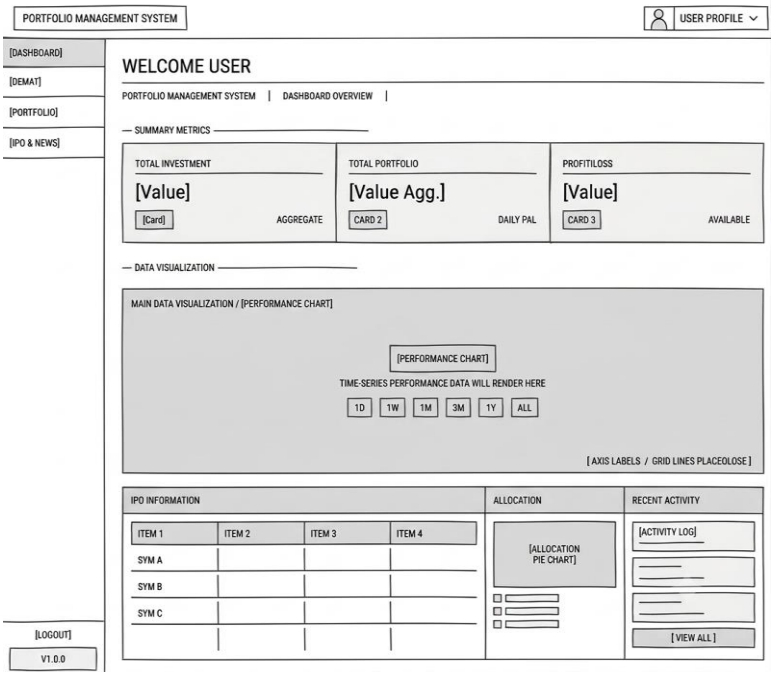


Figure 6: Wireframe of Portfolio Management System

This low-fidelity wireframe maps out a "Portfolio Management System" desktop dashboard using a clean, sketch-like UI design. It features a left-hand navigation sidebar for toggling between core sections like Demat, Portfolio, and IPO info, complete with a logout option at the base. The main interface greets the user with a "WELCOME USER" header positioned right above three financial summary cards tracking investments, portfolio value, and profits. The center is taken up by a large time-series performance chart placeholder with standard historical interval toggles.

7) References

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[8] GeekforGeeks, "GeekforGeeks," 15 January 2026. [Online]. Available: https://www.geeksforgeeks.org/software-engineering/waterfall-model/. [Accessed 15 June 2026].